CH 3 — Diagonalizability

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3.1 Eigenvectors and Diagonalization

Warm Up Question

Let $R : \mathbb{R}^2 \to \mathbb{R}^2$ be reflection in the line $y = 3x$. Determine $R(\vec{v})$

Reflection is a linear map. R can be represented by a 2×2 matrix.

Choose a basis $B = \{e_1, e_2\} \in \mathbb{R}^2$. Build ${}_B[A]_B$

$A = [R]_B = \begin{bmatrix}[R(e_1)]_B & [R(e_2)]_B\end{bmatrix}$. But we don't know $R(e_1), R(e_2)$.

We choose instead the vector $R(v_1) = v_1$, that is on the line $y = 3x$, say $v_1 = \begin{bmatrix}1\\3\end{bmatrix}$

Then choose another vector $R(v_2) = -v_2$, that is perpendicular to the line $y = 3x$, say $v_2 = \begin{bmatrix}-3\\1\end{bmatrix}$

$D = \left\{\begin{bmatrix}1\\3\end{bmatrix}, \begin{bmatrix}-3\\1\end{bmatrix}\right\}$

Now $C = {}_D[R]_D = \begin{bmatrix}[R(v_1)]_D & [R(v_2)]_D\end{bmatrix} = \begin{bmatrix}1 & 0\\0 & -1\end{bmatrix}$

Lesson:

1. Sometimes there may be a "nice" basis for $L : V \to V$ that makes computations easier
2. Dream scenario: want the basis D so that we have $[L]_D = \begin{bmatrix}x & \dots & 0\\ \dots & \dots & \dots\\ 0 & \dots & x\end{bmatrix}$

> 📘 **Definition — Operator**
>
> Let $T : V \to W$ is called a **linear operator** if $\dim V = \dim W$

> 📘 **Definition — Diagonalizability**
>
> Compute the matrix of $L : V \to V$ and obtain the basis D so that we have $[L]_D = \begin{bmatrix}x & \dots & 0\\ \dots & \dots & \dots\\ 0 & \dots & x\end{bmatrix}$
>
> If there is such a basis D, we say L is **diagonalizable** and we say D is a **diagonalizable basis**

Example:

Reflection map $R : \mathbb{R}^2 \to \mathbb{R}^2$ across $y = 3x$ is diagonalizable and $D = \left\{\begin{bmatrix}1\\3\end{bmatrix}, \begin{bmatrix}-3\\1\end{bmatrix}\right\}$ is a diagonalizable basis.

Suppose we want to calcualte $R(\vec{v})$.

$[R(\vec{v})]_D = [R]_D[\vec{v}]_D$

But we want $R(\vec{v}) = [R(\vec{v})]_S, S = \{e_1, e_2\}$

Let ${}_SI_D$ be change for D to S coordinates. Then ${}_SI_D[\vec{x}]_D = [\vec{x}]_S$

We obtain

$$
\begin{aligned}
{}_S I_D[R(\vec{v})]_D &= {}_S I_D[R]_D[\vec{v}]_D \\
[R(\vec{v})]_S &= {}_S I_D[R]_D[\vec{v}]_D \\
[R(\vec{v})]_S &= {}_S I_D[R]_D({}_S I_D)^{-1}{}_S I_D[\vec{v}]_D \\
[R(\vec{v})]_S &= {}_S I_D[R]_D({}_S I_D)^{-1}[\vec{v}]_S
\end{aligned}
$$

So

$$
{}_S[R]_S = {}_S I_D{}_D[R]_D I_S
$$

${}_S I_D = \begin{bmatrix} 1 & -3 \\ 3 & 1 \end{bmatrix}$

$\therefore [R]_S = \begin{bmatrix} 1 & -3 \\ 3 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}\left(\begin{bmatrix} 1 & -3 \\ 3 & 1 \end{bmatrix}\right)^{-1} = \frac{1}{10}\begin{bmatrix} -8 & 6 \\ 6 & 8 \end{bmatrix}$

📘 **Definition — Similarity**

Let $A, B \in M_{n\times n}(\mathbb{F})$, we say A is **similar** to B if $\exists$ invertible matrix P s.t. $A = PBP^{-1}$

If $A \sim B$

1. rank A = rank B
2. nullity A = nullity B
3. det A = det B
4. tr A = tr B

Example:

If B, C are bases for V and if $L : V \to V$ is linear, then ${}_B[L]_B \sim {}_C[L]_C$

since ${}_B[L]_B = {}_B I_C{}_C[L]_S({}_B I_C)^{-1}, P = {}_S I_D$

QUESTION: Given $L : V \to V, \exists D$ s.t. $[L]_D$ is diagonalizable?

3.2

3.3

3.4